Uniwersytet SWPS

Task list no. 3 – We want to work with Quicksort and stacks.

Task 1. Implement the Quicksort algorithm.

Task 2. Compare the performance of Merge Sort and Quick Sort by measuring their execution times for different input sizes ranging from 10 to 10,000 elements. Record the running times for each input size in a table.

Task 3. Implement a stack that supports the following operations:

1. Push: Insert an element onto the top of the stack.
2. Pop: Remove an element from the top of the stack.
3. Display: Print all elements currently in the stack.

The lab report should contain your code along with the results.

Note. ALL PROGRAMS SHOULD COMPILE WITHOUT ERROR AND WORK (i.e. DO EXACTLY WHAT IS DESCRIBED IN THE TASK)!!!

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